

EAST AMATULI STATION LIGHT, AK, USA

WMO: 994081

Lat: 58.915N

Lon: 151.952W

Elev: 33

StdP: 100.93

Time Zone: -9.00 (NAK)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	-12.4	-10.1	-15.7	1.0	-11.1	-14.2	1.1	-9.3	26.8	0.6	23.8	0.7	6.6	320	0.916

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	2.7	17.9	13.5	16.2	12.5	14.8	12.4	14.7	16.6	14.0	15.5	13.4	14.5	4.9	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.0	10.0	14.8	13.4	9.6	14.1	12.8	9.3	13.5	41.0	16.7	39.1	15.7	37.5	14.5	18.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.7	18.7	16.3	DB	-14.4	20.5	2.3	1.7	-16.0	21.7	-17.4	22.7	-18.6	23.6	-20.3	24.8	(4)
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	5.6	0.9	0.7	0.5	2.8	6.7	9.3	11.6	12.6	10.2	6.5	2.7	1.7
	DBStd	5.35	4.97	3.94	3.98	2.71	2.19	2.07	1.93	1.84	1.77	2.34	3.74	3.97
	HDD10.0	1802	282	261	296	215	106	36	4	0	17	108	219	258
	HDD18.3	4666	540	494	554	465	359	271	210	178	243	366	469	516
	CDD10.0	179	0	0	0	1	5	16	53	81	25	0	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	7.4	8.7	8.4	8.2	7.6	7.0	5.8	5.9	6.0	6.8	7.1	8.3	9.1
Precipitation	PrecAvg	1239	129	98	76	70	61	53	72	107	150	142	127	153
	PrecMax	1688	323	228	138	156	115	91	118	192	294	311	332	239
	PrecMin	753	19	28	13	15	20	15	40	31	45	66	22	61
	PrecStd	169	60	42	32	30	29	17	20	41	53	53	74	50
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.7	6.2	6.7	11.8	17.7	19.6	19.2	20.2	15.2	10.7	9.1	7.9
		MCWB	2.5	5.3	5.2	7.2	10.1	15.2	14.7	14.7	11.4	7.8	8.6	7.3
	2%	DB	6.8	5.7	6.0	7.7	13.0	15.6	17.3	18.1	13.7	10.1	8.4	7.1
		MCWB	4.2	4.8	4.9	5.4	7.2	10.8	13.6	14.2	11.8	8.3	8.1	6.4
	5%	DB	6.2	5.2	5.6	6.4	10.9	13.5	15.6	16.4	13.0	9.6	7.9	6.6
		MCWB	4.8	4.4	4.5	5.0	6.4	10.4	12.8	12.9	11.7	8.5	7.5	5.9
	10%	DB	5.7	4.8	5.0	5.5	9.4	11.8	14.3	15.1	12.4	9.1	7.2	6.0
		MCWB	4.3	3.9	4.0	4.5	6.5	9.6	12.6	12.9	11.2	8.1	6.8	5.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.1	5.4	5.8	7.4	10.7	13.2	15.6	16.2	13.5	10.1	9.1	7.9
		MCDB	6.2	5.5	6.4	11.4	17.2	15.1	18.1	18.4	13.6	10.1	9.1	8.0
	2%	WB	5.4	4.9	5.3	6.3	8.9	11.9	14.5	15.1	12.8	9.5	8.4	7.2
		MCDB	5.8	5.0	5.8	7.8	11.9	13.2	16.4	17.0	13.0	9.5	8.4	7.3
	5%	WB	5.0	4.3	4.7	5.4	8.1	10.8	13.8	14.3	12.4	9.0	7.9	6.5
		MCDB	5.4	4.6	5.3	5.9	9.6	12.2	15.4	16.0	12.6	9.1	8.0	6.7
	10%	WB	4.4	3.8	4.1	4.8	7.7	10.1	13.1	13.8	11.9	8.5	7.2	5.8
		MCDB	5.0	4.2	4.6	5.3	8.9	11.2	14.5	15.1	12.2	8.7	7.4	6.2
Mean Daily Temperature Range	5% DB	MDBR	3.1	3.1	3.0	2.6	2.5	2.8	2.8	2.7	2.1	2.1	3.0	3.2
		MCDBR	2.1	2.4	2.4	3.0	5.5	6.0	5.1	4.7	2.8	1.9	2.7	2.7
		MCWBR	2.2	2.8	2.6	2.0	2.6	2.9	2.7	2.5	2.2	2.3	2.6	2.9
	5% WB	MCDBR	2.0	2.7	2.6	2.9	4.0	4.6	4.6	3.9	2.1	2.0	2.3	2.3
		MCWBR	2.4	3.0	2.6	2.0	2.4	3.0	3.0	2.7	2.1	2.3	2.4	2.8
Clear-Sky Solar Irradiance	taub	0.272	0.314	0.327	0.368	0.382	0.368	0.370	0.364	0.347	0.324	0.289	0.280	
	taud	2.269	2.347	2.375	2.298	2.319	2.450	2.491	2.502	2.499	2.394	2.208	2.059	
	Ebn at Noon	558	712	816	835	848	868	857	837	792	687	506	365	
	Edh at Noon	43	66	85	109	114	102	96	89	76	62	44	37	
All-Sky Solar Radiation	RadAvg	0.34	0.91	2.23	3.60	4.82	5.26	4.73	3.80	2.54	1.27	0.49	0.22	
	RadStd	0.05	0.15	0.34	0.46	0.49	0.47	0.46	0.45	0.35	0.13	0.08	0.02	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (12 neighbors)		+0.84	+2.34	+2.18	N/A	N/A	N/A	-225	-287	N/A	N/A	

Nomenclature: See separate page